



WEEK 3: ASSIGNMENT 3

Question 1:

Distance protection is preferred over graded time-lag overcurrent protection for HV and EHV transmission lines because

- a) it operates faster.
- b) it is simpler.
- c) it is less expensive.
- d) All of these

Answer: (a)

Question 2:

A 3-phase, 11 kV power system has the following phase currents during an unsymmetrical fault: $I_a=100\angle 0^\circ$ A, $I_b=50\angle -120^\circ$ A, $I_c=30\angle 120^\circ$ A. The positive sequence current component is

- a) 20 A
- b) 60 A
- c) 80 A
- d) 180 A

Answer: (b)

Solution:

Positive sequence current component is,

$$I_1 = \frac{1}{3}(I_a + aI_b + a^2I_c) = \frac{1}{3}(100\angle 0 + 1\angle 120 \times 50\angle -120 + 1\angle 240 \times 30\angle 120)$$
$$I_1 = \frac{1}{3}(100 + 50 + 30) = 60 \text{ A}$$

Question 3:

The purpose of backup protection is to

- a) increase the speed of operation.
- b) increase the reach of protection.
- c) eliminate blind spots in protection.
- d) provide protection in case of failure of primary protection.

Answer: (d)

Question 4:

Which zone of a distance relay is most affected by the transient overreach phenomenon?

- a) Zone 1
- b) Zone 2
- c) Zone 3
- d) None of these

Answer: (a)

Question 5:

A short transmission line having an impedance of $(2+j5) \Omega$ is protected by a reactance relay. The CT ratio is 500/1 A and PT ratio is 132 kV/110 V. If the relay is adjusted to just operate for a dead short-circuit at the end of line section, then the set value of relay is

- a) 2.944
- b) 2.083
- c) 2.517
- d) 3.614

Answer: (b)

Solution:

$$X_{relay} = X \times \frac{CT \text{ ratio}}{PT \text{ ratio}} = 5 \times \frac{\left(\frac{500}{1}\right)}{\left(\frac{132000}{110}\right)} = 2.083$$

Question 6:

Which protective relay is commonly used for the protection of very long EHV transmission lines?

- a) Overcurrent relay with extremely inverse characteristics
- b) Percentage differential relay
- c) Mho relay
- d) Reactance relay

Answer: (c)

Question 7:

Due to current infeed from the remote end of a transmission line, the distance relay located at the local end tends to

- a) overreach.
- b) underreach.
- c) Both (a) and (b)
- d) operate normally without being affected.

Answer: (b)

Question 8:

Fig. 1 shows the single line diagram of a 400 kV three-terminal transmission line system. A three-phase fault occurs on the line section near the bus C. The magnitude and direction of currents from each bus are shown in the Fig. 1. In addition, the impedance of line section connected between bus A and T-point, between bus B and T-point and between bus C and T-point is $25\angle 80^\circ$, $15\angle 85^\circ$ and $12\angle 75^\circ$, respectively. Then, the apparent impedance seen by relay located at bus A is

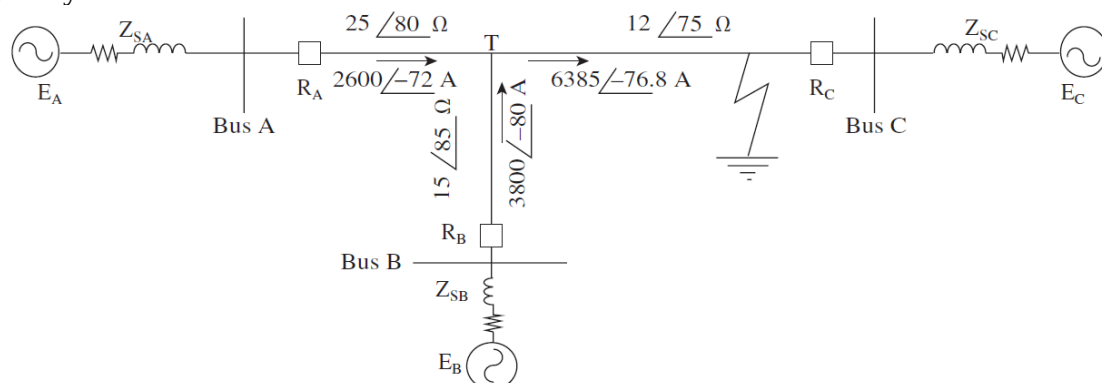


Fig. 1 Single line diagram of the three-terminal transmission line

- a) $44.3\angle 74.7^\circ \Omega$
- b) $44.3\angle 44.4^\circ \Omega$
- c) $54.3\angle 74.7^\circ \Omega$
- d) $54.3\angle 44.4^\circ \Omega$

Answer: (c)

Solution:

$$Z_A = \frac{V_A}{I_A} = \frac{12\angle 75^\circ \times 6385\angle -76.8^\circ + 25\angle 80^\circ \times 2600\angle -72^\circ}{2600\angle -72^\circ} = 54.3\angle 74.7^\circ \Omega$$

Question 9:

Overreaching of a distance relay caused by the decaying DC component can be minimized by using

- a) electromechanical relay.
- b) solid-state relay.
- c) digital relay.
- d) None of these

Answer: (c)

Question 10:

A 132 kV transmission line has an impedance of $(4 + j16) \Omega$. This line is protected by mho type distance relay with CT ratio = 1000/1 A and PT ratio = 132 kV/110 V. The zone-1 of the reactance relay covers 80% of the line length, zone-2 covers 150% of the line length. Then, the first and second zone setting of reactance relay are

- a) 20.67Ω and 10Ω , respectively
- b) 10.67Ω and 20Ω , respectively
- c) 10Ω and 20.67Ω , respectively
- d) 20Ω and 10.67Ω , respectively

Answer: (b)

Solution:

$$Z_{line} = 4 + j16 \Omega, \quad X_{line} = 16 \Omega$$

Zone-1 setting (80% of line)

$$X_{Zone-1} = 0.8 \times X_{line} \times \frac{CT \text{ ratio}}{PT \text{ ratio}} = 0.8 \times 16 \times \frac{\left(\frac{1000}{1}\right)}{\left(\frac{132000}{110}\right)} = 10.67 \Omega$$

Zone-2 setting (150% of line)

$$X_{Zone-2} = 1.5 \times X_{line} \times \frac{CT \text{ ratio}}{PT \text{ ratio}} = 1.5 \times 16 \times \frac{\left(\frac{1000}{1}\right)}{\left(\frac{132000}{110}\right)} = 20 \Omega$$

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